

```

In[*]:=  $\alpha = 0.0005499651141443689$ ;

In[*]:=  $T = 491.535$ ;

In[*]:=  $w_0 = \sqrt{(2 * \text{Pi} / T)^2 + (\alpha / 2)^2}$ ;
      raíz cuadrada número pi

In[*]:=  $\tau = 2 * 0.05 * (1 - 0.058772338) * 6.863155982898358^{(-11)} * ((1.5 * 0.0383) / 0.0422^2)$ ;

In[*]:= (*El 2 de aquí abajo es mb, se podría cambiar*)

In[*]:=  $I_1 = 2 * 0.0383 * (0.05^2 + (2 / 5) * 0.00955^2) + (1 / 3) * 2 * 0.05^2$ ;

In[*]:=  $w_1 = 2 * \text{Pi} / T$ ;
      número pi

In[*]:=  $\theta[t\_]:= (\tau / (I_1 * w_0^2)) * (1 - E^{-(\alpha * t) / 2} * (\text{Cos}[w_1 * t] + (\alpha / (2 * w_1)) * \text{Sin}[w_1 * t]))$ 
      número e coseno seno

In[*]:= p = Plot[2 *  $\theta[t]$  * 318.54, {t, -100, 8000},
      representación gráfica
      BaseStyle -> {FontFamily -> "Calibri", FontSize -> 14}, PlotStyle ->
      estilo base familia de tipo de letra tamaño de tipo de letra estilo de representación
      {Blend[{Blue, White}], Thick}, PlotRange -> {{-100, 4000}, {-0.2, 8}}, Frame -> True,
      mezcla azul blanco grueso rango de representación marco verdad
      FrameLabel -> {Style["t / s", Bold, 16], Style["2L $\theta$  / rad cm", 16, Bold]},
      etiqueta de marco estilo negrita estilo negrita
      PlotLegends -> {"Curva 2L $\theta$  (t)"};
      leyendas de representación

In[*]:= dataTHETAmin = {{0, 0}, {1 * T, 0.4}, {2 * T, 0.9},
      {3 * T, 1.2}, {4 * T, 1.4}, {5 * T, 1.9}, {6 * T, 2.1}, {7 * T, 2.3}};

In[*]:= dataTHETAmx = {{0.5 * T, 7.9}, {1.5 * T, 7.4}, {2.5 * T, 7.1},
      {3.5 * T, 6.5}, {4.5 * T, 6.2}, {5.5 * T, 6}, {6.5 * T, 5.6}, {7.5 * T, 5.4}};

In[*]:= plotdatamin = ListPlot[dataTHETAmin,
      representación de lista
      PlotStyle -> {Black, Thickness[0.003]}, PlotLegends -> {"Datos experimentales"};
      estilo de representación negro grosor leyendas de representación

In[*]:= plotdatamax = ListPlot[dataTHETAmx, PlotStyle -> {Black, Thickness[0.003]}];
      representación de lista estilo de representación negro grosor

In[*]:= Needs["ErrorBarPlots`"];
      necesita

```

 **General:** ErrorBarPlots` is now obsolete. The legacy version being loaded may conflict with current functionality. See the Compatibility Guide for updating information.

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In[*]:= dataerrmin = ErrorListPlot[
      {{0, 0}, ErrorBar[0, 0.102833734]}, {{1 * T, 0.4}, ErrorBar[4.606, 0.102833734]},
      {{2 * T, 0.9}, ErrorBar[9.212, 0.102833734]}, {{3 * T, 1.2},
      ErrorBar[13.819, 0.102833734]}, {{4 * T, 1.4}, ErrorBar[18.425, 0.102833734]},
      {{5 * T, 1.9}, ErrorBar[23.032, 0.102833734]}, {{6 * T, 2.1},
      ErrorBar[27.638, 0.102833734]}, {{7 * T, 2.3}, ErrorBar[32.244, 0.102833734]}},
      AxesOrigin -> {0, 0}, PlotStyle -> {Black, Thickness[0.002]}];
      origen de ejes estilo de representación negro grosor

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In[*]:= dataerrmax =
  ErrorListPlot[{{0.5 * T, 7.9}, ErrorBar[2.303, 0.102833734]}, {{1.5 * T, 7.4},
    ErrorBar[6.909, 0.102833734]}, {{2.5 * T, 7.1}, ErrorBar[11.516, 0.102833734]},
    {{3.5 * T, 6.5}, ErrorBar[16.122, 0.102833734]}, {{4.5 * T, 6.2},
    ErrorBar[20.728, 0.102833734]}, {{5.5 * T, 6}, ErrorBar[25.335, 0.102833734]},
    {{6.5 * T, 5.6}, ErrorBar[29.941, 0.102833734]},
    {{7.5 * T, 5.4}, ErrorBar[34.548, 0.102833734]}}],
  AxesOrigin -> {0, 0}, PlotStyle -> {Black, Thickness[0.002]};
  |origen de ejes |estilo de repre... |negro |grosor

In[*]:= Show[p, plotdatamin, plotdatamax, dataerrmin, dataerrmax]
  |muestra

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Out[*]=

